

Research overview

One systematic equity strategy: how its simulated headline was audited and rebuilt, and how results are checked.

HEADLINE FIGURES: HISTORICAL BACKTEST · SELECTED CONFIGURATION. NO LIVE OR PAPER-TRADING RESULTS ARE SHOWN.

3.04× Backtest ending multiple, Jun 2024 – Jul 2026, about 70% a year	1.84 Backtest Sharpe ratio, daily returns, no risk-free rate subtracted	−19.1% Backtest maximum drawdown	780 Simulated trades, \$1M simulated start
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Selected configuration result. Configuration selection and evaluation overlap in this research window, so it is not independent out-of-sample evidence. Deflated Sharpe ratio: 0.85.

What this is

The subject is a long-only, market-hours systematic equity strategy that trades short-horizon relative-value dislocations between single stocks and statistically formed peer groups, with a portfolio-level risk governor.

Around it sits a research platform: minute-resolution history for 5,000+ U.S. equities, a research replay harness, a transaction-cost model, a verification layer, and a trading engine with a checksummed release pipeline.

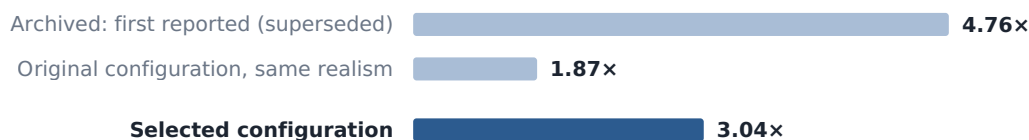
The platform has progressed from research and simulation into live trading with real capital through broker-integrated execution. Every performance figure in this brief is a historical backtest. No live or paper-trading results are shown.

How the headline was reached

In late September 2026 I reported a two-year walk-forward backtest (models fitted on past data, tested on later unseen data) that ended at 4.76×, now archived as superseded.

An audit I directed in October found hindsight in the list of tradable stocks: one 2026 snapshot covered every earlier date. Costs were a flat charge, with every order filled in full. I had the simulation rebuilt on a point-in-time universe: the tradable list as of each date under a listing and minimum-liquidity rule. That cost breadth: fewer trades. Modeled fills (spread, square-root market impact, partial fills) took the original configuration to 1.87×. I accepted the lower figure because the evidence behind it is more credible.

The selected configuration keeps both. It revises how the liquidity screen is measured as of each date, and the level of the minimum-liquidity requirement, which restores breadth. It ends at 3.04×. With the revised screen and the original requirement, the figure is 2.45×.



Ending value of \$1 after about two years, \$1M simulated start, in three historical backtests. The first bar's window ends about a week before the others. The last bar is not a further audit step: its liquidity requirement was chosen afterward, on this same window.

How results are checked

- **Pre-registration.** For 16 of 18 studies in the October 2026 research sweep, a written entry stating the question preceded the report in version control, by under three hours. None is registered externally.
- **Placebo.** Placebo versions of the revised liquidity screen, with its information removed, ended in the same range as the real one. I credit that revision with no skill of its own.
- **Era split and family-wise test.** Within the sweep, rules are chosen on the earlier half of the sample and judged on the later half. One test controlling false discoveries across the whole family (Reality Check and SPA, Romano-Wolf step-down) covered 69 proposed improvements and 44 placebo twins. One survived, an effect already known, and it failed on the later half alone. The twins behaved as a null.
- **Trial accounting.** My first reported backtest deflated its Sharpe ratio for 100 trials. A recount found about 540 runs in 62 families. The selected configuration's deflated Sharpe ratio (the probability that its true Sharpe beats the best expected by luck) is 0.85 against 240 full-window simulations. The conventional bar is 0.95.
- **Parity proofs.** Before any study ran, the unmodified research copy of the simulator had to reproduce the reference simulation to the dollar. It did. The patched build the studies then ran on made the same trades and ended about fifty dollars away, so each study is compared with a reference run on its own build.

Execution and capacity

With order sizes scaled to the starting balance, the selected configuration's modeled cost per round trip rises from about 30 basis points at \$100k to about 40 at \$1M and about 55 at \$5M, and its ending multiple falls as the balance rises. I quote no single capacity figure. The impact coefficient is estimated from public tape data, not measured on my own orders.

Author

I am an undergraduate at the University of Miami, graduating in December 2026 with a B.A. in Mathematics (Probability & Statistics track) and a B.S. in Finance & Accounting. Implementation uses extensive AI-assisted development; the research questions, the methodology and the adoption decisions are mine.

Limits

- The headline is a historical backtest: one implementation of one two-year window. Configurations were chosen on that window, so it holds no untouched holdout. The S&P 500 rose about 42% over that window on a price basis; the simulated book can use leverage, and market exposure has not been separated from the result.
- I chose the minimum-liquidity requirement after testing five settings on that window. Every revised setting beat the original after costs, and several settings are within noise of one another.
- A two-year compounding multiple moves about $\pm 0.5\times$ on small changes: one small rule change took the same book from $3.04\times$ to $2.50\times$. The step from $2.45\times$ to $3.04\times$ is about that size.
- A small fraction of trades accounts for more than half of net profit. The final mark is about 18% below the peak.
- Dividends and later-delisted stocks are outside the simulation ledger. Measured separately on the original configuration, they add a few percent of net profit. Financing cost is not charged.
- The work has not yet had independent outside review.

Selected material presented for educational and professional purposes. Proprietary signal definitions, parameters, portfolio construction rules and execution logic are intentionally omitted. All performance figures are historical backtests unless labeled otherwise; no live or paper-trading results are shown. Historical and simulated results do not guarantee future performance. Nothing here is an offer or investment advice.